

Introduction to stability and control problems

ELEC0447 - Analysis of electric power and energy systems

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November 25, 2025

Overview

1. The concept of stability
2. Why do we need stability?

Why to control power systems?

- ▶ Technical requirements: power system devices are designed so as to operate within well-defined "tolerance regions"
 - ▶ around nominal values of voltage V_n : 1 ± 0.1 pu in Europe
 - ▶ around nominal value of frequency f_n : 50 ± 0.2 Hz in Europe (in steady state)
 - ▶ within the P - Q capabilities of devices
 - ▶ under the current limits of lines and transformers
- ▶ Large/persistent deviations from nominal values could lead to
 - ▶ damages and safety problems (e.g. high voltage)
 - ▶ cascading phenomena
 - ▶ service interruptions

Exogeneous threats

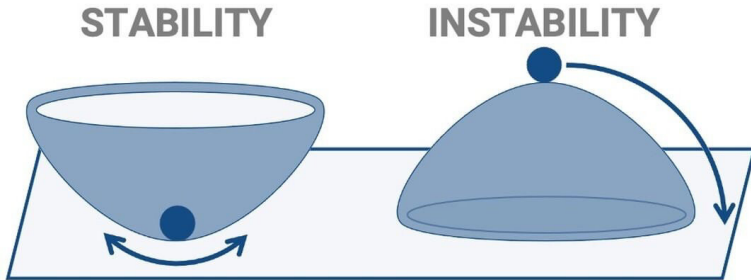
- ▶ Sudden disturbances, such as line or generator tripping
- ▶ Fast variations of the *net load* (cf. Duck curve)
 - ▶ the net load refers to the load "seen" by the transmission system, i.e. the load minus the non-controllable dispersed generation
- ▶ weather conditions, such as storms, which can impact the generation of renewable energy sources (RES) (e.g. wind turbines' cut-out speed)

The concept of stability

The background of the slide features a white upper half and a teal lower half. The teal section is composed of two large triangles meeting at a point at the bottom center, with a smaller, darker teal triangle at the very bottom.

General concept

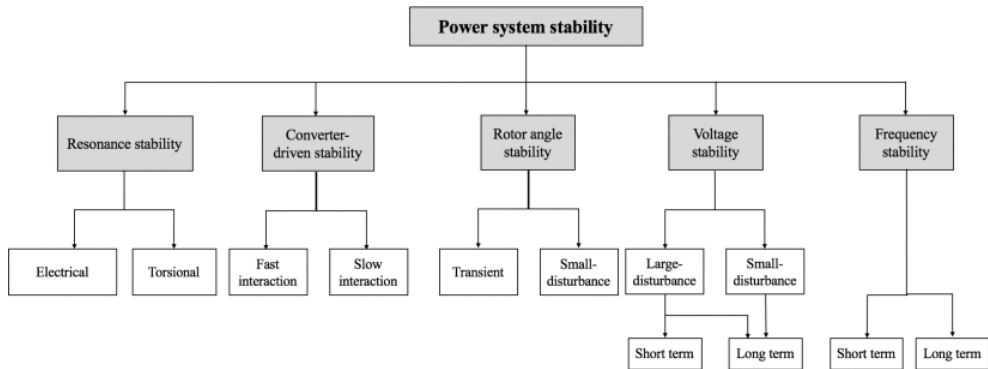
If a system has the property that it will get back into the equilibrium state again after moving away from its equilibrium state, then it is stable. [1]



In power systems I

Power system stability is the ability of an electric power system, for a given initial operating condition, to regain a state of operating equilibrium after being subjected to a physical disturbance, with most system variables bounded so that practically the entire system remains intact. [2]

In power systems II



Why do we need stability?

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large triangles, one on the left and one on the right, point towards the center. These triangles overlap at the bottom, creating a darker teal triangular shape. The top half of the slide is a plain white background where the text is located.

Key points

- ▶ We often take electricity as a simple commodity.
- ▶ But the electric power system is one of the most complex and largest man-made system.
- ▶ The chances of system failures are very high taking into account the impact of external factors and rapid changes in system's state.
- ▶ However, power systems are very reliable (operated 24h/24h 7d/7d and only a few hours of power outages per year!).
- ▶ But when instabilities occur, it can lead to blackouts with huge financial and societal consequences.

Tokyo 1987

- Unexpected load increase and presence of constant power devices (air conditioners) led to a voltage collapse.
- **8 GW lost** and **2.8M customers impacted** [3, 4]

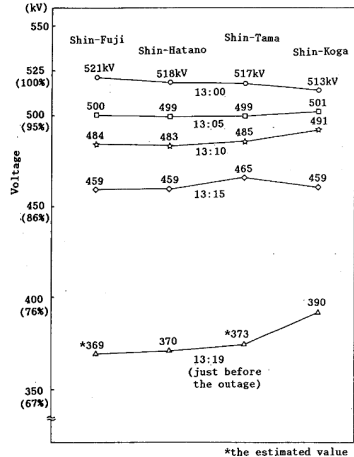
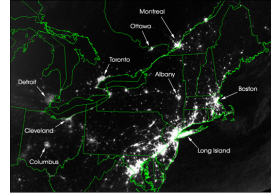


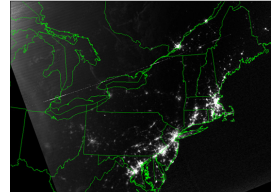
Fig.3 Voltage Drops at Main Substations

Canada/Northeast USA 2003

- ▶ Initial problem: not enough reactive power reserve. Hot weather and large consumption led to transmission lines overloading and eventually sagging into trees, further deteriorating the initial problem.
- ▶ A cascading event caused the tripping of hundreds of lines and generating units.
- ▶ **63 GW lost and 50M customers impacted and Estimated cost above \$5 Billion**
- ▶ Watch <https://www.youtube.com/watch?v=nd3teNgUq8E>,
<https://www.youtube.com/watch?v=KciAzYfXNwU>



August 14, 2003 • 9:29 p.m. EDT • About 20 hours before blackout

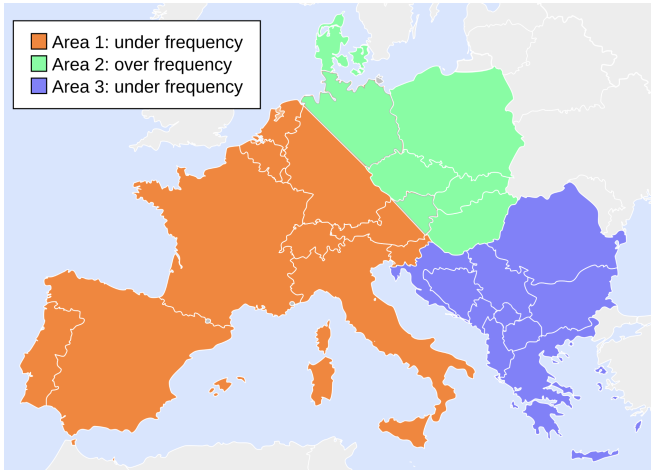


August 15, 2003 • 9:14 p.m. EDT • About 7 hours after blackout

Europe 2006 I

- ▶ Disconnection of a transmission line in Germany for the transport of a ship approved by the local TSO.
- ▶ The local TSO approved to advance the disconnection later that day, but the commercial flows remained unchanged.
- ▶ Some lines were critically loaded because of the line disconnection and a fast increase of load consumption led to a cascading event.
- ▶ European interconnected network has been split into 3 islands.
- ▶ Watch <https://www.youtube.com/watch?v=A30DdnsICuw>
- ▶ **14.5 GW lost and 15M customers impacted [5]**

Europe 2006 II



Brazil 2023

- ▶ False operation of a relay protection system led to a 500kV line disconnection.
- ▶ The Energy Management System did not operate properly.
- ▶ **19 GW lost**

Conclusion

- ▶ Every time a black-out happened, lessons have been learned, and new rules have been put in place.
- ▶ Nonetheless, due to the system complexity, new complex phenomena occur that power system engineers try to understand.
- ▶ In the following, we'll dive into the mechanisms of voltage instability.

References I

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